ACTIVITY 1.4 Observation, inference and prediction

SKILLS

- Critical and creative thinking
- Questioning and predicting
- Evaluating

1 Examine the following images. For each one, describe what you observe, what you can infer from the picture, and any predictions you can make. The first one has been done for you.

a

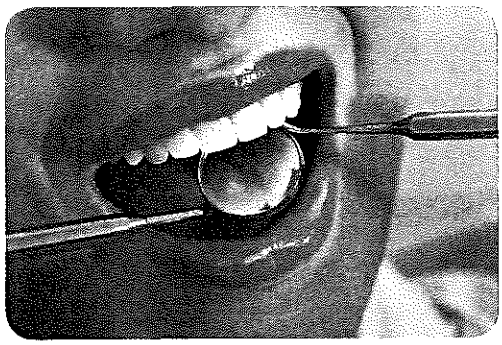


Fig 1.10

Observations: A dentist is examining a patient's teeth.

Inference: The patient has either gone for a routine check-up or has a problem with their teeth.

Prediction: The patient may require treatment or just a clean.

b



Fig 1.11

Observations: _____

_____

Inferences: _____

_____

Predictions: _____

_____

c

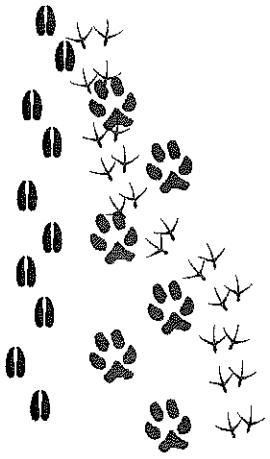


Fig 1.12

Observations: _____

_____

Inferences: _____

_____

Predictions: _____

_____

d

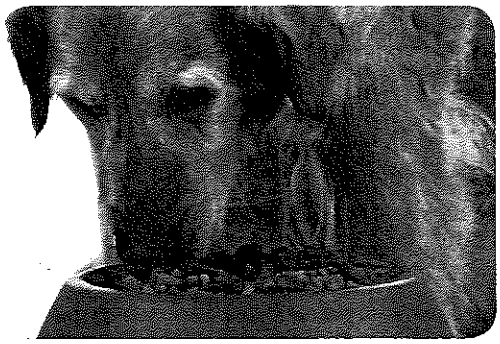


Fig 1.13

Observations: _____

_____

Inferences: _____

_____

Predictions: _____

_____

e

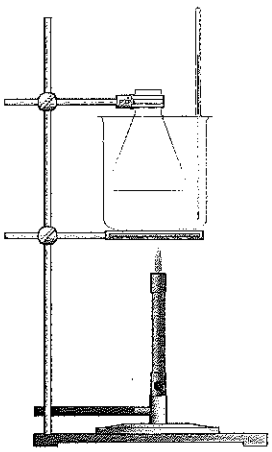


Fig 1.14

Observations: _____

_____

Inferences: _____

_____

Predictions: _____

_____

f

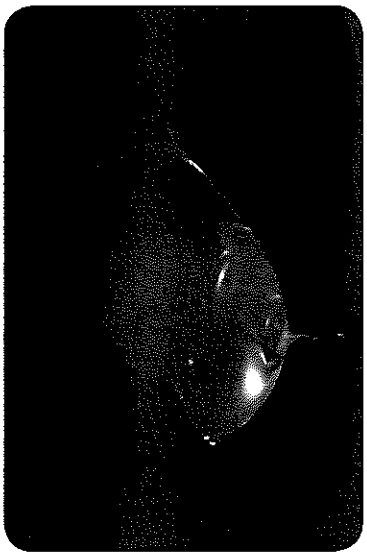


Fig 1.15

Observations: _____

_____

Inferences: _____

_____

Predictions: _____

_____

2 Scientists often have to list observations, come up with inferences and make predictions. Sometimes scientists cannot take direct observations, but must rely on computers or robotic probes to make measurements and relay the information back. In 1976, some strange observations were recorded from the planet Mars. NASA's *Viking 1* spacecraft was circling the planet, snapping photos of possible landing sites for its sister ship *Viking 2*. Many people were stunned to see the shadowy figure shown in Figure 1.16, which was snapped at a place called Cydonia on the Martian surface. This caused the start of many rumours.

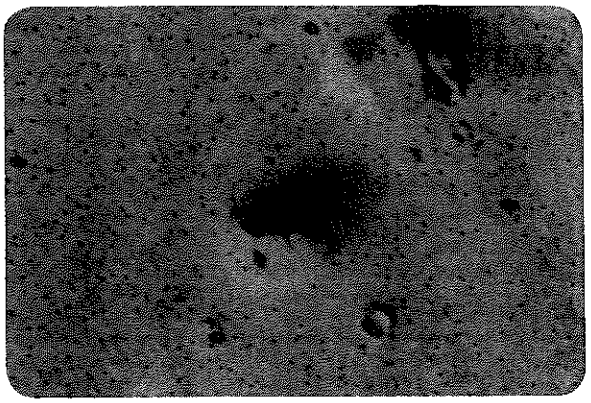


Fig 1.16

a List four observations from this photo taken from the *Viking* probe.

b Make two different inferences based on this photo.

c Predict what a future space probe might find on the surface of Mars.

d Explain how a prediction is different from an inference.